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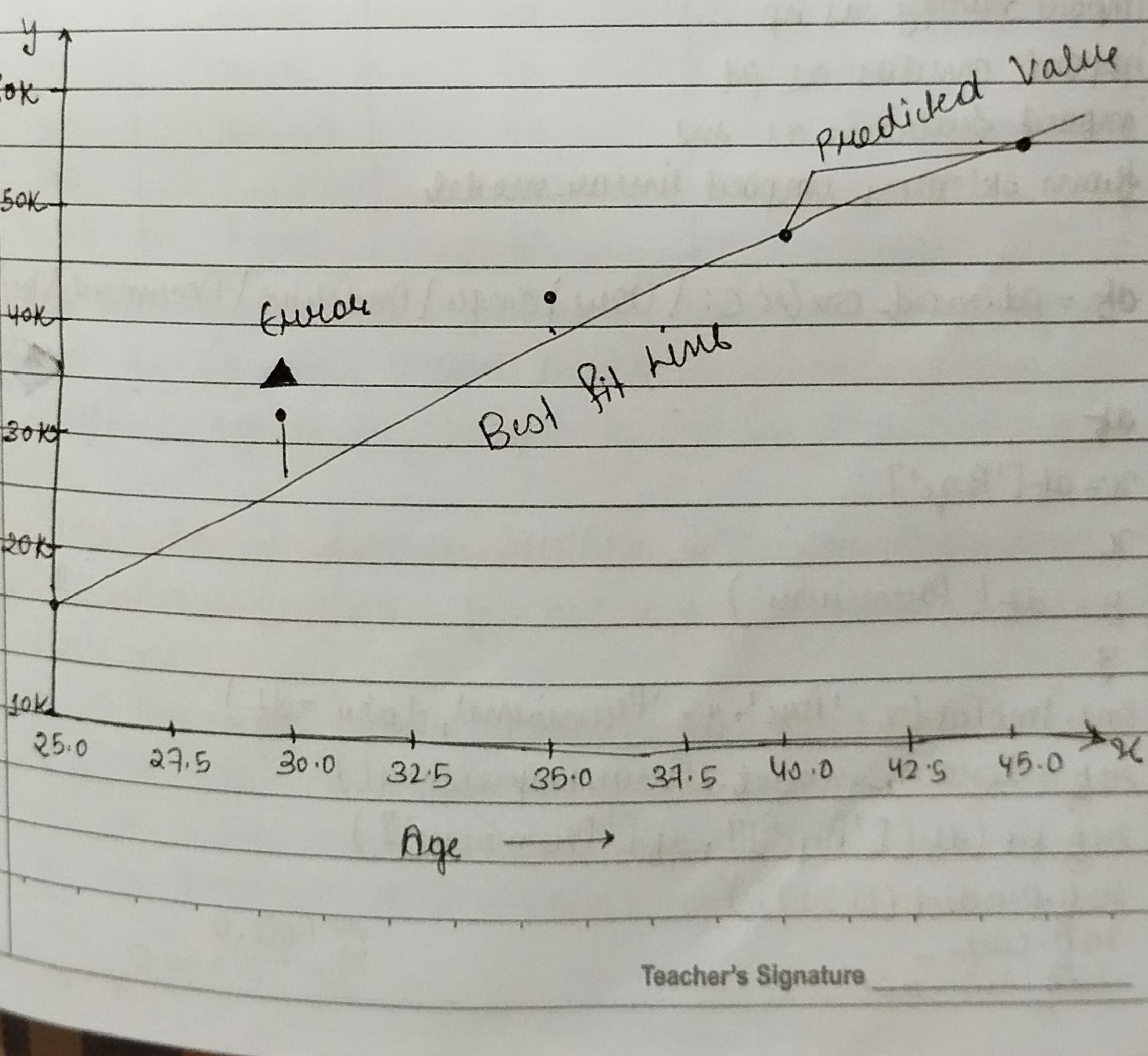
★ Linear Regression

Linear Regression is a machine learning algorithm based on Supervised learning.

It is a Statistical method that is used for predictive analysis. Linear Regression makes predictions for continuous / Real or numeric variables such as cost, age, sales, temperature, product price, etc.

- Best-Fit line



Teacher's Signature _____

- **Independent & Dependent Variable**

$y = mx + c$ (linear equation)

where,

$y \rightarrow$ dependent variable

$x \rightarrow$ Independent variable

$m \rightarrow$ slope / gradient / coefficient

$c \rightarrow$ Intercept

Premium = m* age + c

- **Code** (Term_Insurance.csv)

```
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn import linear_model

df = pd.read_csv(r'C:\Users\singh\OneDrive\Documents\Term_Insurance.csv')

df
x = df['Age']
x
y = df('Premium')
y
sns.lmplot(x = 'Age', y = 'Premium', data = df)
reg = linear_model.LinearRegression()
reg.fit(df[['Age']], df['Premium'])
reg.predict([[21]])
reg.coef_                    // 1780.0
reg.intercept_               // -23500
```

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- verify linear Equation with code

let's $y = mx + c$

where, $m = 1780$, $x = 25$ and $c = -23500$

$y = (1780) * (25) + (-23500)$

$y = 37880 - 23500$

$y = 13880$ (verified)

★ Term Insurance Dataset

Age	height	weight	Premium
25	162.56	70	18000
30	172.72	95	38000
35	167.64	78	38000
40		110	60000
45	167.48	85	70000

We will find out Premium whose age is 27 height is 165.56 and weight is 60

whose age is 60, height is 165.40 and weight is 80.

- Formula of linear Multiple Variable Regression.

linear Equation : $y = m_1 * x_1 + m_2 * x_2 + m_3 * x_3 + c$

such as:

$y \rightarrow$ dependent variable (Premium)

$x_1 \rightarrow$ independent variable (Age)

$x_2 \rightarrow$ independent variable (Height)

$x_3 \rightarrow$ independent variable (Weight)

$c \rightarrow$ Intercept

$m_1, m_2, m_3 \rightarrow$ Slope / Coefficient / gradient

Teacher's Signature _____

- Code

```
import pandas as pd
from sklearn import linear_model

df = pd.read_csv(r'C:\Users\singh\OneDrive\Documents\Insurance.csv')
df

mean_height = df.Height.mean()
mean_height

df.Height = df.Height.fillna(mean_height)
df

reg = linear_model.LinearRegression()
reg.fit(df[['Age','Height','Weight']], df['Premium'])

reg.coef_                    // array [[2150.26052416, -248.45851674,
reg.intercept_               // -16827.013154824956      312.65291961]]

reg.predict([[27, 165.56, 60]])      //array (18854.40430838)
reg.predict([[60, 165.10, 80]])      // array (96180.35091498)
```

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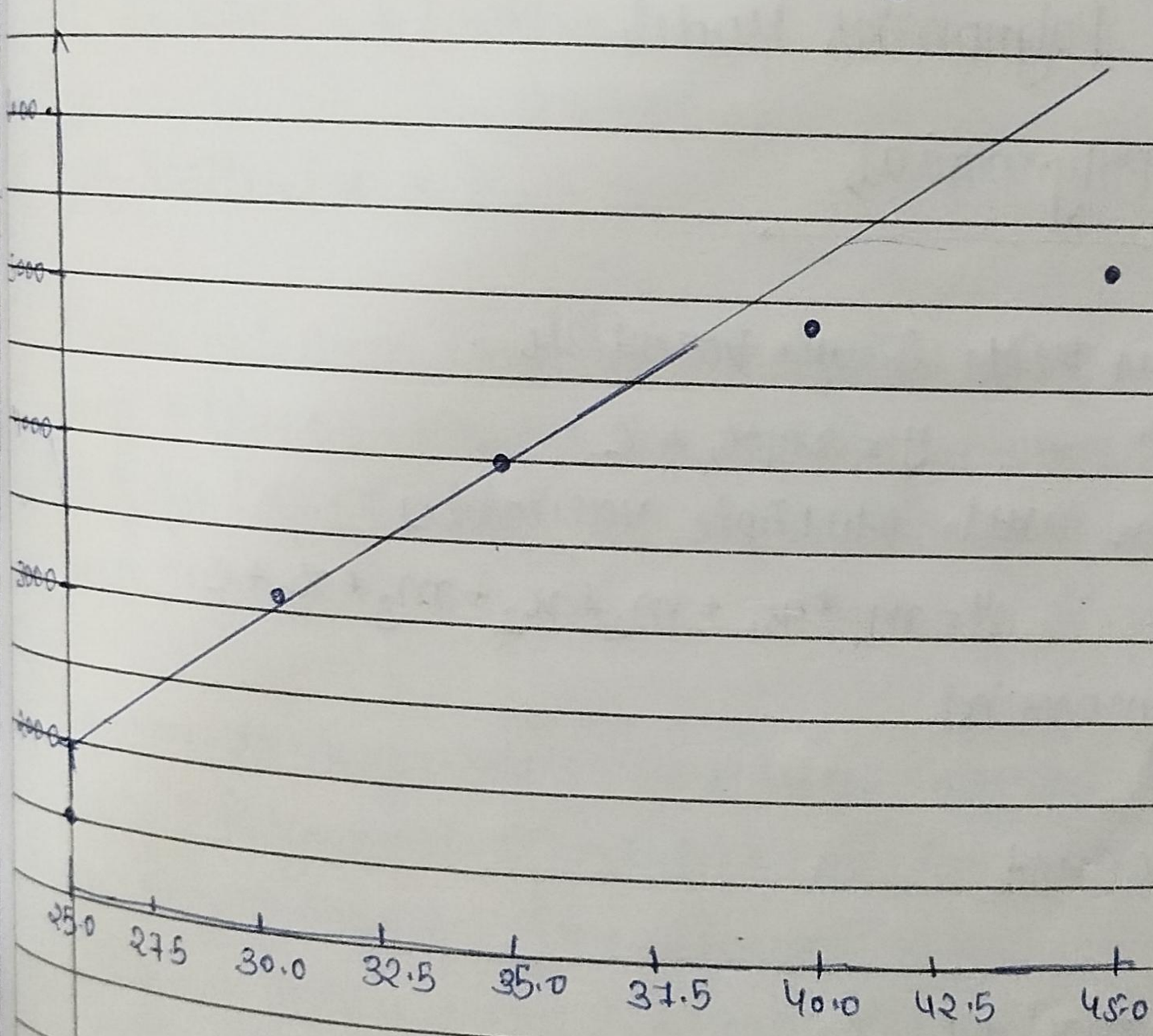
• Verify with linear Equations

w.k.t $y = m_1x_1 + m_2x_2 + m_3x_3 + c$

$y = (2150.26052416)(27) + (-248.45851374)(165.56)$
$+ (312.65291961)(60) + (-16827.013154824$
$y = 18854.404$ (verified) 956)

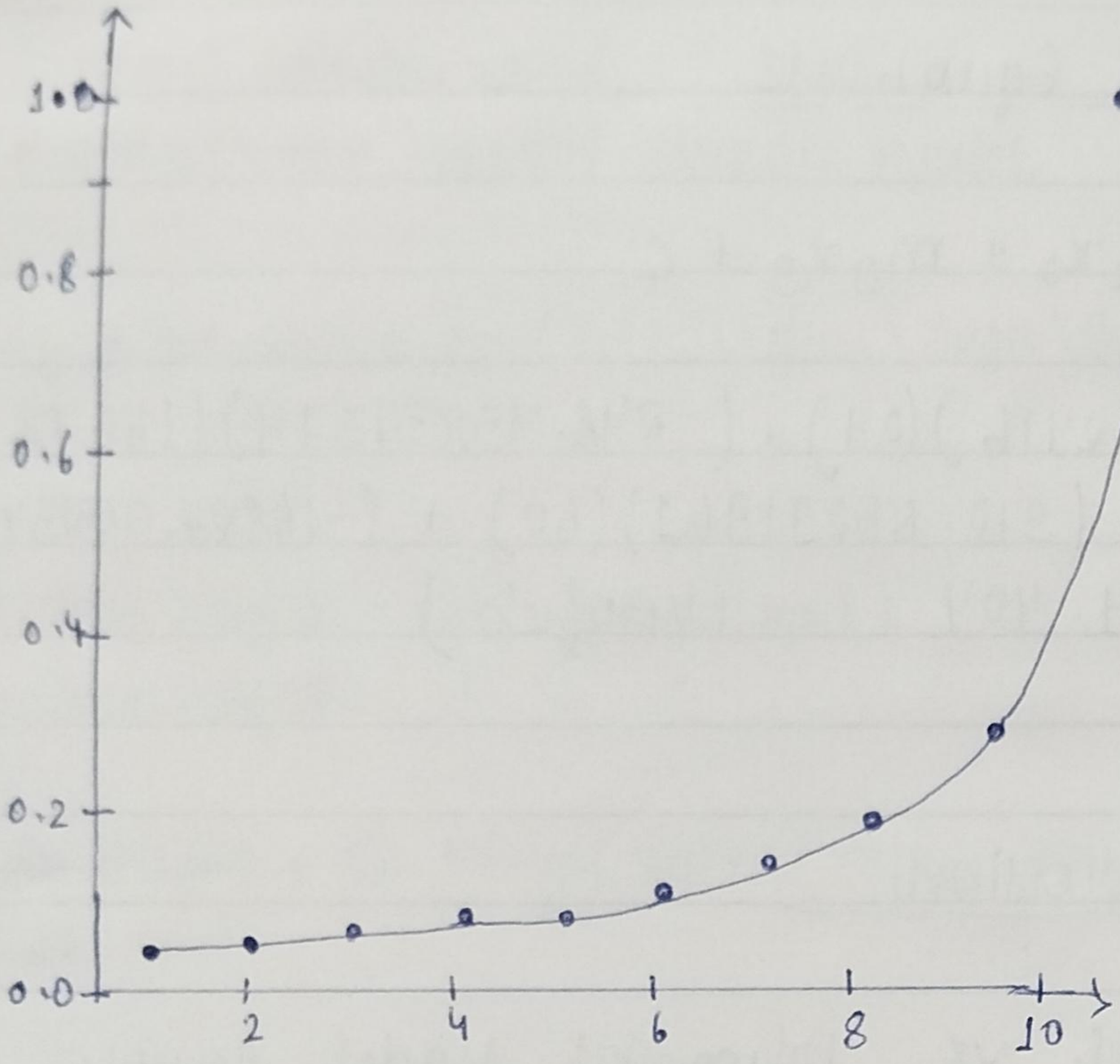
★ Polynomial Regression

Simple Linear Model v/s Polynomial Model graph



Simple Linear Model

Teacher's Signature _____



Polynomial Model

★ ① Degree of Polynomial

Linear Equation with Single variable

$$y = m_1 x_1 + c$$

Linear Equation with Multiple variables

$$y = m_1 * x_1 + m_2 * x_2 + m_3 * x_3 + c$$

Degree of Polynomial

0 Degree

$$y = \text{const}$$

1 Degree

$$y = mx^1 + c$$

2 Degree

$$y = ax^2 + bx + c$$

Generalized Polynomial

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \text{----} + a_n x^n$$

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- Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("C:\Users\singa\OneDrive\Documents\Position_Salaries.csv")

df
x = df.iloc[:, 1:2].values

x
y = df.iloc[:, 2].values

y
plt.scatter(x, y)


from sklearn import linear_model
reg = linear_model.LinearRegression()
reg.fit(df[['Level']], df['Salary'])
reg.predict([[6.5]])                                    // 330378.7878


from sklearn.preprocessing import PolynomialFeatures
poly = PolynomialFeatures(degree=2)
x_poly = poly.fit_transform(x)
reg2 = linear_model.LinearRegression()
reg2.fit(x_poly, y)
reg2.predict(poly.fit_transform([[6.5]]))  // 189498.106
```

Teacher's Signature _____

★ Logistic Regression (Binary classification)

Logistic Regression is a machine learning algorithm based on supervised learning.
It is a statical method that is used for predicting probability of target variables.
Logistic Regression makes probability for classification Problems that are discrete in nature.

- Binary Classification — Ex: yes or no, dead or alive
- Multiclass-classification — Ex: onion or potato or tomato

- Logistic Regression Real Life Example
  - Heart Attack prediction
  - Tumour prediction
  - Credit-Card Fraud detection
  - Spam detection

- Term Insurance Dataset

Age	Bought Insurance
21	no
48	yes
32	yes
41	yes
20	no
35	yes
20	no
23	no

Bought Insurance
0
1
1
1
0
1
0
0

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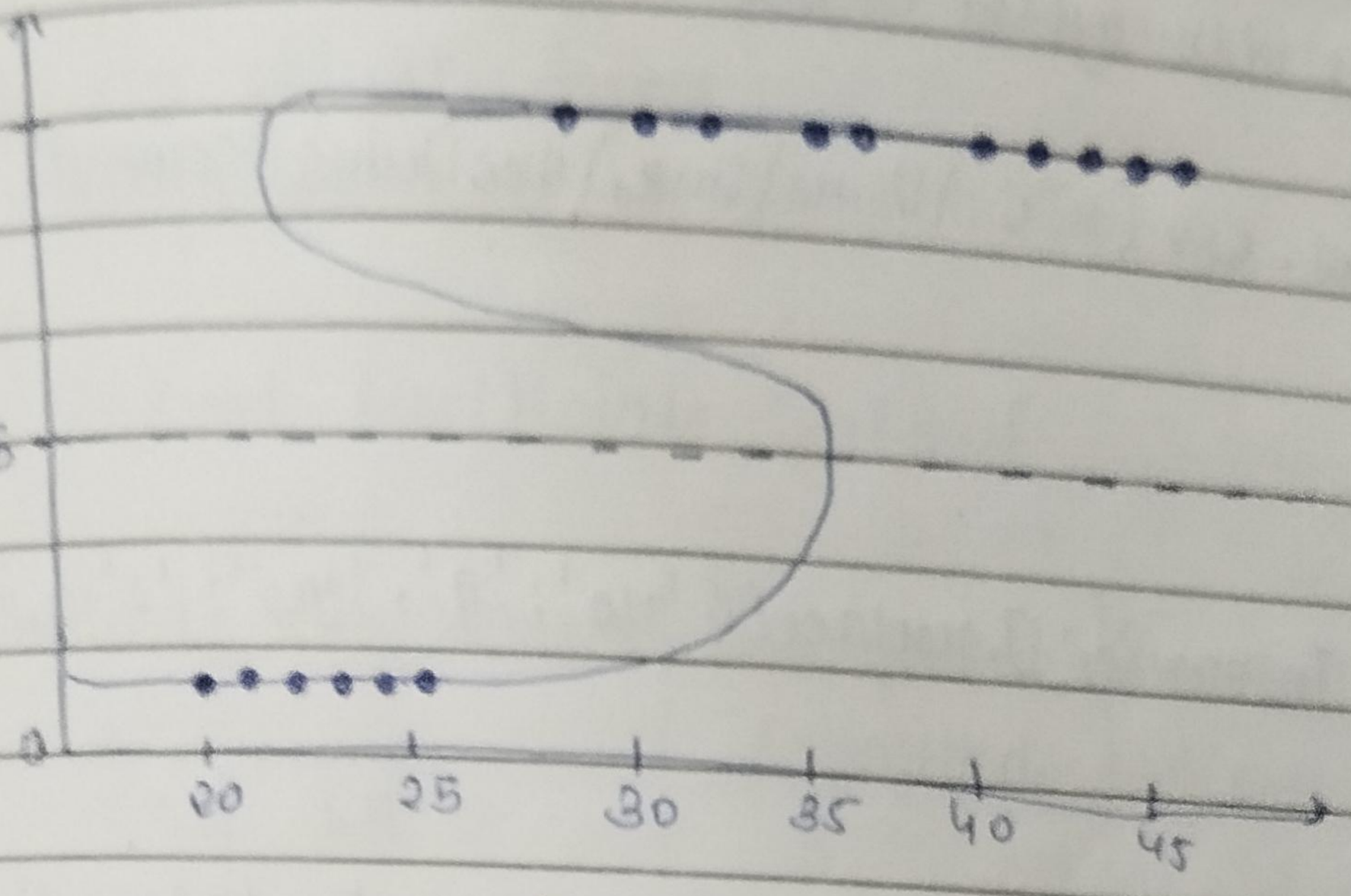
Graphs

Logistic Regression Graph



Formula of Sigmoid function will be

$$y = 1 / 1 + e^{-x}$$

converts input (x) into range 0 to 1.

Such as,

$y \rightarrow$ dependent variable (Bought insurance)

$x \rightarrow$ independent variable (Age)

$e \rightarrow$ Euler's constant ≈ 2.71828

Teacher's Signature _____

- Code

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv(r"C:\Users\Singe\OneDrive\Documents\TERM-INSURANCE
                 - DATASET.csv")
```

```
df
```

```
df['Bought Insurance'].replace({'no':'0','yes':'1'},inplace=True)
df
```

```
plt.scatter(x='Age', y='Bought Insurance', data=df)
```

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(df[['Age']],
                                   df['Bought Insurance'], test_size=0.4
```

```
x_train
len(x_train)
x_test
len(x_test)
```

```
from sklearn.linear_model import LogisticRegression
lr = LogisticRegression()
lr.fit(x_train, y_train)
lr.predict(x_test)
```

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* Multiclass Classification Real examples
identifying animals types from photos
identifying vehicles types
news categorizations

• Sample Student Multiclass Dataset

Hours-Studied	Attendance	Assignment Submitted	Previous Score	Extra-Curricular	Final Result
2	65	3	45	NO	Low
3	70	4	50	NO	Med
4	75	5	55	Yes	Med
5	80	6	60	Yes	Med
6	85	7	65	Yes	High
7	90	8	70	Yes	High
8	92	9	75	Yes	High

• Code

```
import Pandas as pd
df = pd.read_csv(r"C:\Users\Singh\Downloads\Student_Multiclass_dataset.csv")
df['Extracurricular'].replace({'NO':0, 'Yes':1}, inplace=True)
df['Final-Result'].replace({'Low':1, 'Medium':2, 'High':3}, inplace=True)
df
```

Teacher's Signature _____

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(df[['Hours_Studied',
'Attendance', 'Assignments_Submitted', 'Previous_Score',
'Extracurricular']], df['Final_Result'],
test_size=0.2)

x_train
len(x_train)
x_test
len(x_test)

from sklearn.linear_model import LogisticRegression
lr = LogisticRegression()
lr.fit(x_train, y_train)
lr.predict(x_test)
lr.score(x_test, y_test)

import seaborn as sns
sns.pairplot(df[['Hours_Studied', 'Attendance', 'Assignments_
Submitted', 'Previous_Score', 'Extracurricular',
'Final_Result']], hue='Final_Result')
```

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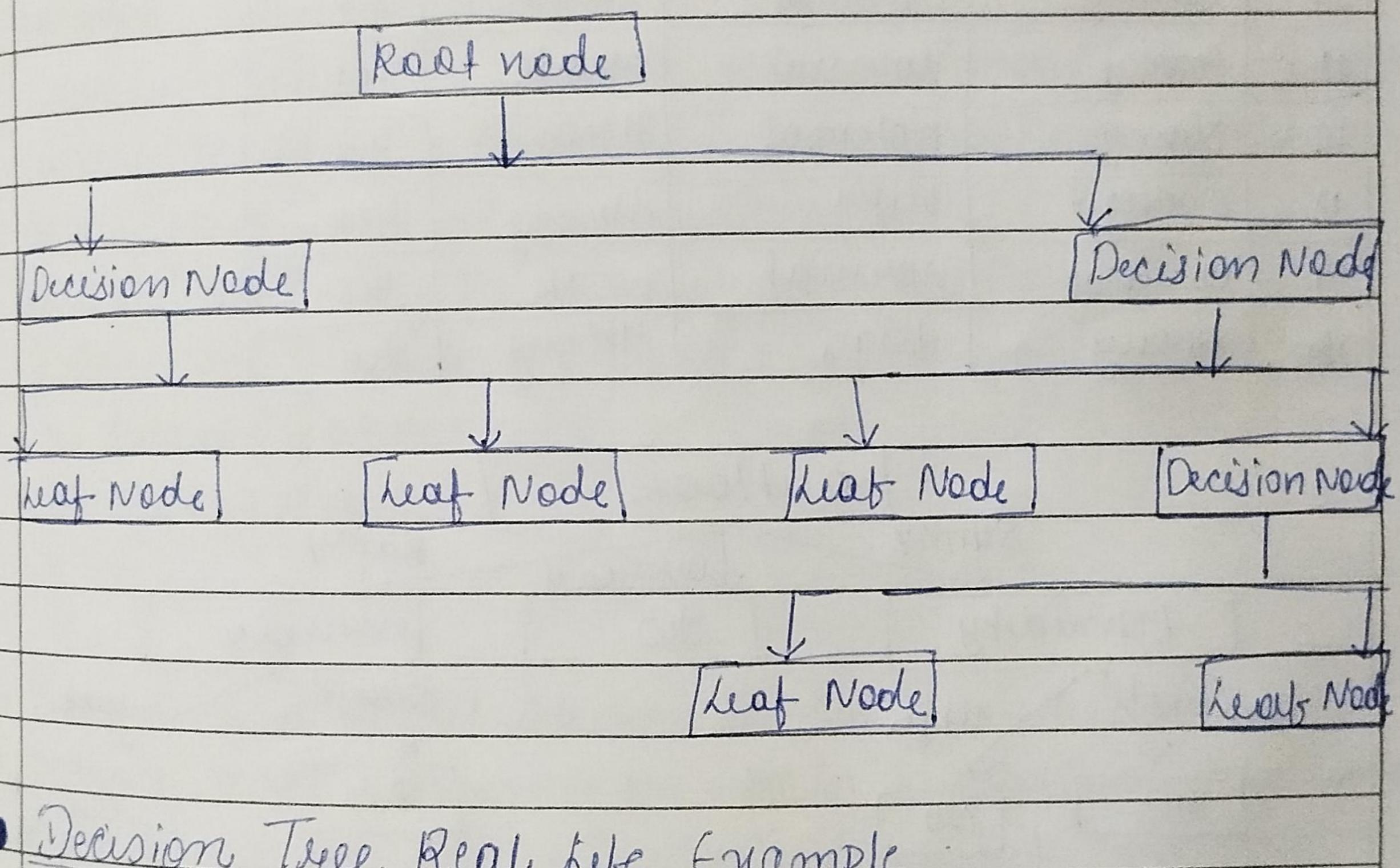
★ Decision Tree in ML

Decision Tree is a machine learning algorithm based on supervised learning, that can be used for both regression and classification problems.

The goal is to build up a classifier model that can predict the class or value of the target variable.



Decision Tree Real Life Example

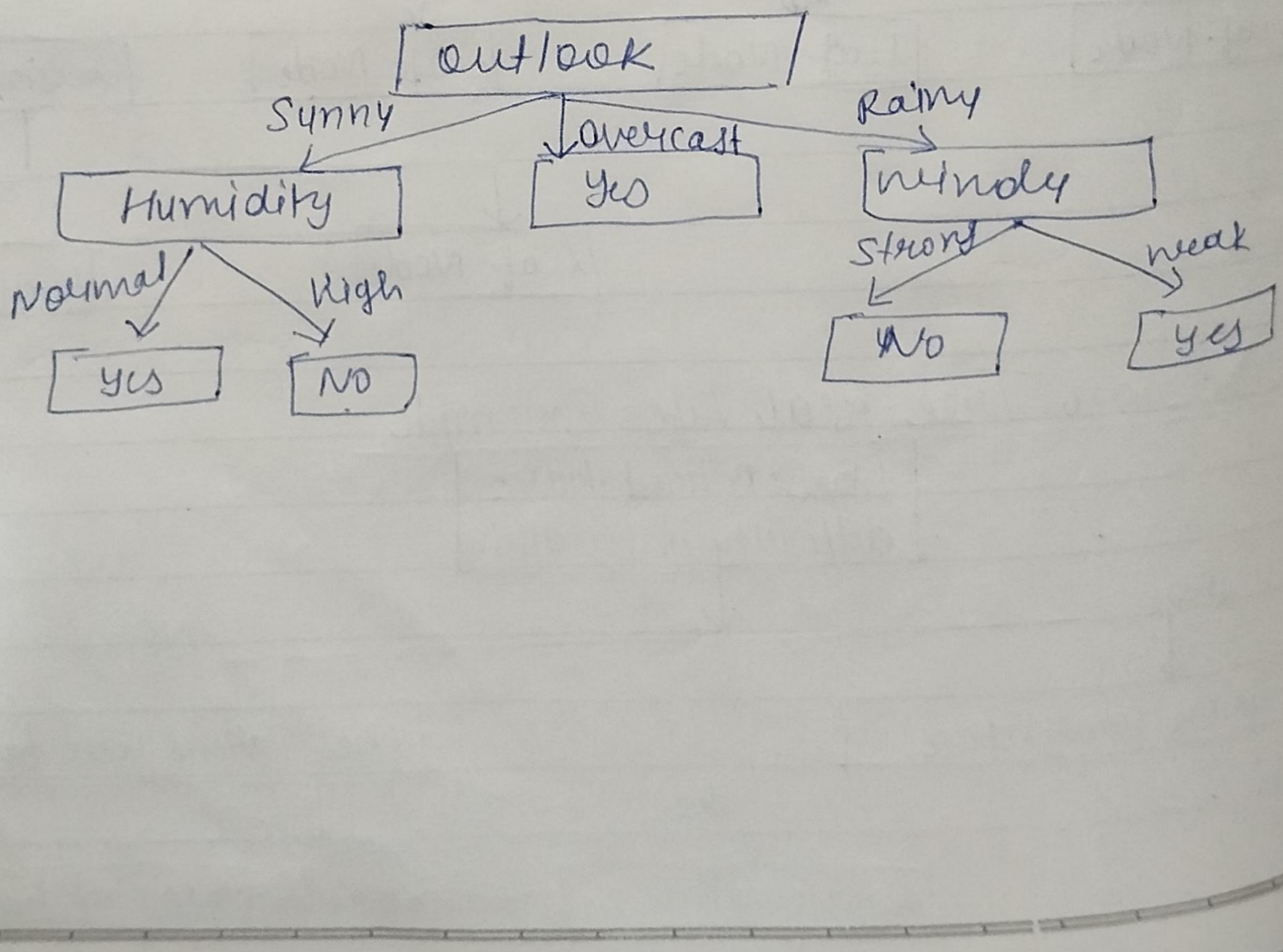
Teacher's Signature

Medical Field

• How we create a Decision Tree ?

	outlook	humidity	wind	Play
1	outlook	high	weak	no
2	sunny	high	strong	no
3	sunny	high	weak	yes
4	overcast	high	weak	yes
5	rainy	normal	weak	yes
6	rainy	normal	strong	no
7	rainy	normal	weak	yes
8	overcast	high	weak	no
9	sunny	normal	weak	yes
10	sunny	normal	weak	yes
11	rainy	normal	strong	yes
12	sunny	high	strong	yes
13	overcast	high	weak	yes
14	overcast	normal	weak	yes
15	rainy	high	strong	no

Sunny (2)
Overcast (0)
Rainy (1)
no (0)
Yes (1)
weak (1)
Strong (0)
high (0)
normal (1)



Code

```
import pandas as pd
dataset = pd.read_csv("C:\Users\Singh\Downloads\decision-tree-dataset.csv")
dataset
from sklearn.preprocessing import LabelEncoder
Outlook = LabelEncoder()
Humidity = LabelEncoder()
Wind = LabelEncoder()
Play = LabelEncoder()
dataset['Outlook'] = Outlook.fit_transform(dataset['Outlook'])
dataset['Humidity'] = Humidity.fit_transform(dataset['Humidity'])
dataset['Wind'] = Wind.fit_transform(dataset['Wind'])
dataset['Play'] = Play.fit_transform(dataset['Play'])
features_col = ['Outlook', 'Humidity', 'Wind']
x = dataset[features_col]
y = dataset.Play
x
y
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)
from sklearn.tree import DecisionTreeClassifier
classifier = DecisionTree(classifier(criterion='gini')
classifier.fit(x_train, y_train)
classifier.predict(x_test)
x_test
y_test
classifier.score(x_test, y_test)
```

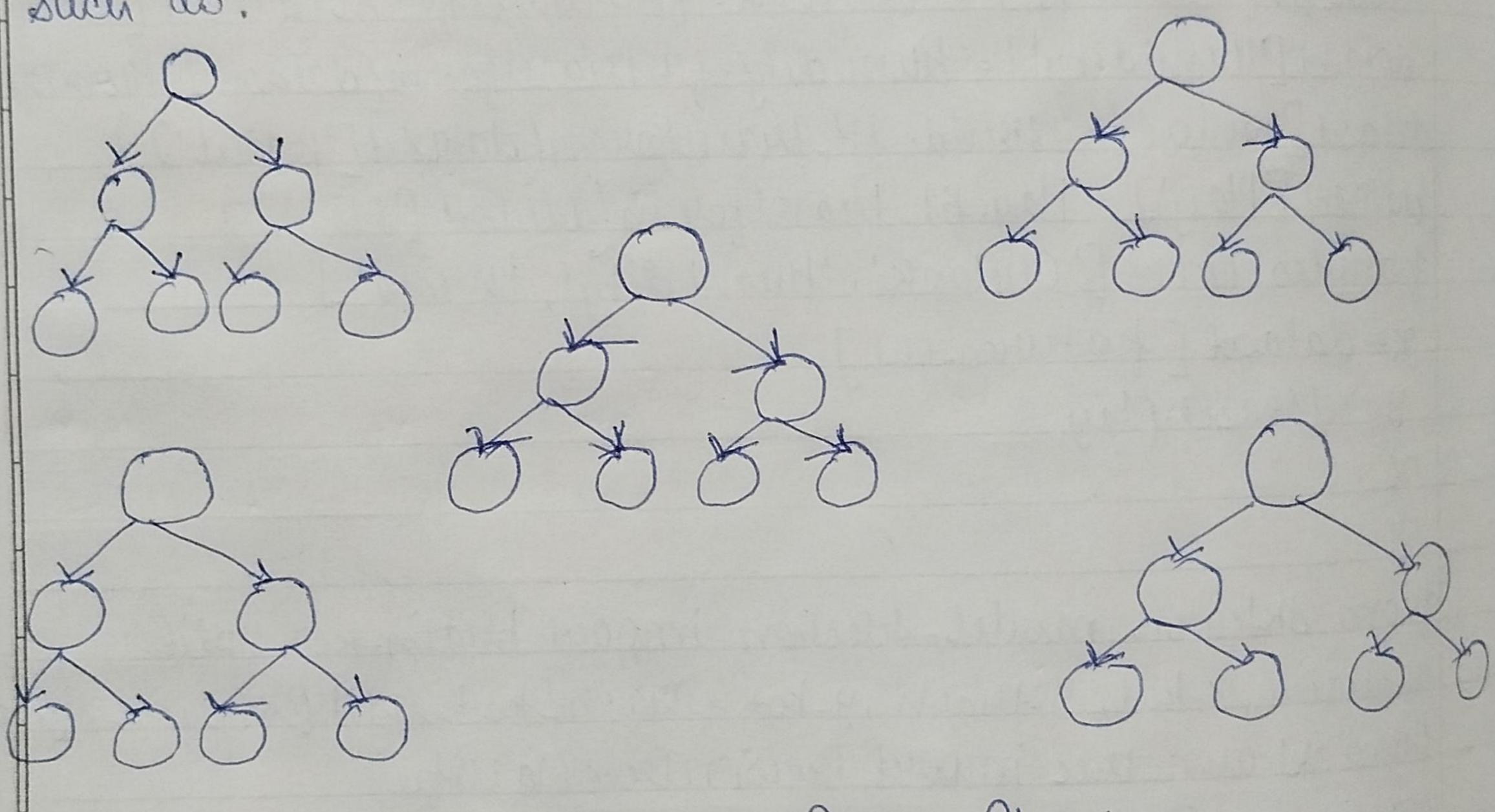
```
from sklearn import tree
tree.plot_tree(classifier)
```

★ Random Forest Algorithm

Random Forest is a machine learning algorithm based on supervised learning that can be used for both regression and classification problems.

It is a collection of multiple random decision trees, which is called the forest.

such as:



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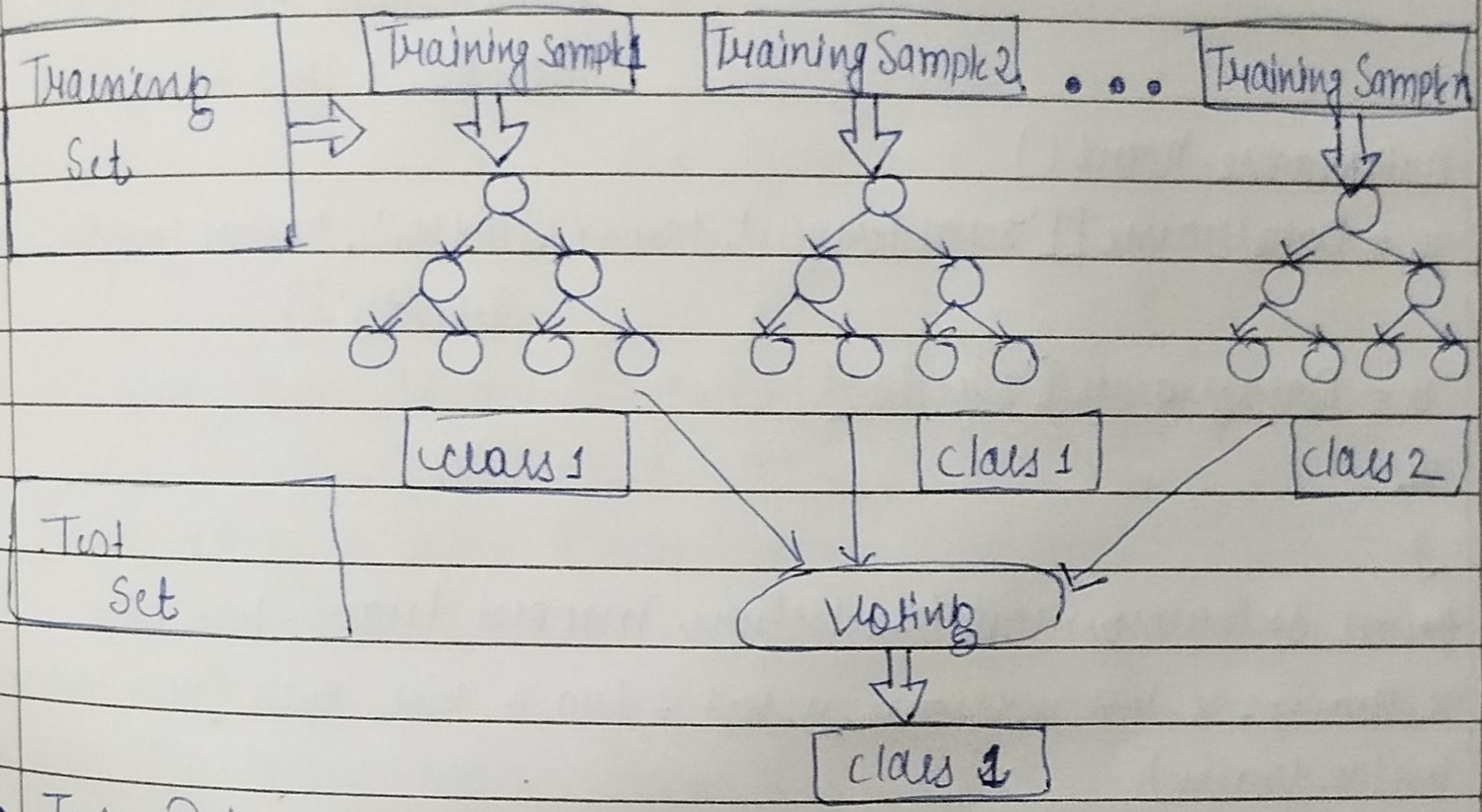
- Real life Example of RFA

1. Financial Fraud detection
2. Healthcare Diagnosis
3. E-commerce Recommendations

- How does the Algorithm works ?



- Iris Dataset

SepalLength(cm)	SepalWidth(cm)	PetalLength(cm)	PetalWidth(cm)	Species
6.8	3.2	5.9	2.3	(2) Iris-Virginica
6.9	3.1	5.1	2.3	"
4.9	3.0	1.4	0.2	(0) Iris-Setosa
5.6	3.0	4.5	1.5	(1) Iris-Versicolor
4.8	3.1	1.6	0.2	Iris-Setosa
5.8	2.8	5.1	2.4	Iris-Virginica
7.2	3.6	6.1	2.5	"
5.1	3.5	1.4	0.3	Iris-Setosa
4.7	3.2	1.6	0.2	"
6.6	3.0	4.4	1.4	Iris-Versicolor

Teacher's Signature _____

- Code

```
from sklearn import datasets
iris = datasets.load_iris()
Dataframe.head()
iris
import pandas as pd
Dataframe = pd.DataFrame({'sepal length' : iris.data[:,0],
                          'sepal width' : iris.data[:,1],
                          'petal length' : iris.data[:,2],
                          'petal width' : iris.data[:,3],
                          'Species' : iris.target})
Dataframe.head()
x = Dataframe[['sepal length', 'sepal width', 'petal length', 'Petal
                                                    width']]
y = Dataframe['Species']
x
y
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3)
len(x_train)
len(x_test)
from sklearn.ensemble import RandomForestClassifier
clf = RandomForest(classifier(n_estimators = 100, criterion='gini')
clf.fit(x_train, y_train)
clf.predict(x_test)
x_test
clf.score(x_test, y_test)
clf.predict([[3.7, 2, 6]])
feature_imp = pd.Series(clf.feature_importances_, index = iris.
                        feature_names).sort_values(ascending
                                                   False)
feature_imp
```

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Naive Bayes Classification Algorithm

- Naive Bayes classifier is a machine learning algorithm based on supervised learning, that can be used for solving classification problems.

- It is Probabilistic classifier, which means it predicts on the basis of the probability of an object.

Such as:

- Tossing a coin Probability
  $S = \{H, T\}$
- Tossing two coins Probability
  $S = \{HH, HT, TH, TT\}$
- Throwing a dice Probability
  $S = \{1, 2, 3, 4, 5, 6\}$

- **Why the Name Naive Bayes ?**
- It comprises of two words Naive and Bayes:

Naive: It's called naive because it makes the assumption that all attributes are independent of each other.
such as: Fruit = {green, oval, sweet} ⇒ watermelon

Bayes: It's called bayes because it depends on the Principle of Bayes Theorem.

Teacher's Signature _____

What is Conditional Probability?

such as: playing card example
4 Suits = Hearts, clubs, Diamonds, Spades
13 cards = Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King
Each of 13 cards has all 4 suits.

if you pick a card from the deck, can you guess the probability of getting a king given the card is a spade?

Total Spade = 13
king = 1
P(king / Spade) = 1/13 = 0.076
P(A|B) = Probability of A occurring given that B has already occured.

- ## What is Bayes Theorem?

The naive Bayes classifier works on the principle of conditional probability, as given by the Bayes Theorem.

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Where,
A, B = Event
P(A|B) = Probability of A given that B is true.
P(B|A) = Probability of B given that A is True.
P(A) = Probability of event A.
P(B) = Probability of Event B

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- Examples of Bayes Theorem.

Let's take Spade & King card. example:

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

$$P(King|Spade) = \frac{P(Spade|King) * P(King)}{P(spade)}$$

given: $P(Spade|King) = \frac{1}{4}$, $P(King) = 1/13$

, $P(Spade) = 1/4$

$$P(King|Spade) \Rightarrow \frac{(1/4) * (1/13)}{(1/4)} \Rightarrow \frac{1}{52} \times \frac{4}{1} \Rightarrow \frac{1}{13}$$

$\Rightarrow 0.076$

- Where Naive Bayes is Used?

- Face Recognition
- Weather Forecast
- News Categorization

- Code

```
import pandas as pd
dataset = pd.read_csv(r"C:\Users\singh\Downloads\Social_Networks_Ads.csv")
dataset
x = dataset.iloc[:, [1,2,3]].values
x
y = dataset.iloc[:,-1].values
y
```

Teacher's Signature _____

```
from sklearn.preprocessing import LabelEncoder
lb = LabelEncoder()
x[x[:,0] = lb.fit_transform(x[:,0])
x
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0
len(x_train)
len(x_test)
from sklearn.naive_bayes import GaussianNB
model = GaussianNB()
model.fit(x_train, y_train)
model.predict(x_test)
x_test
model.score(x_test, y_test)
```

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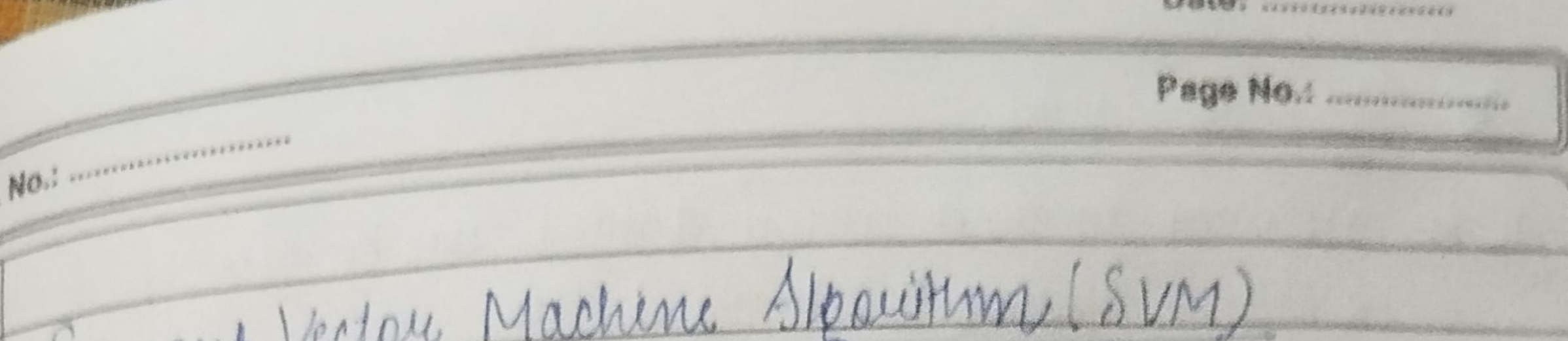
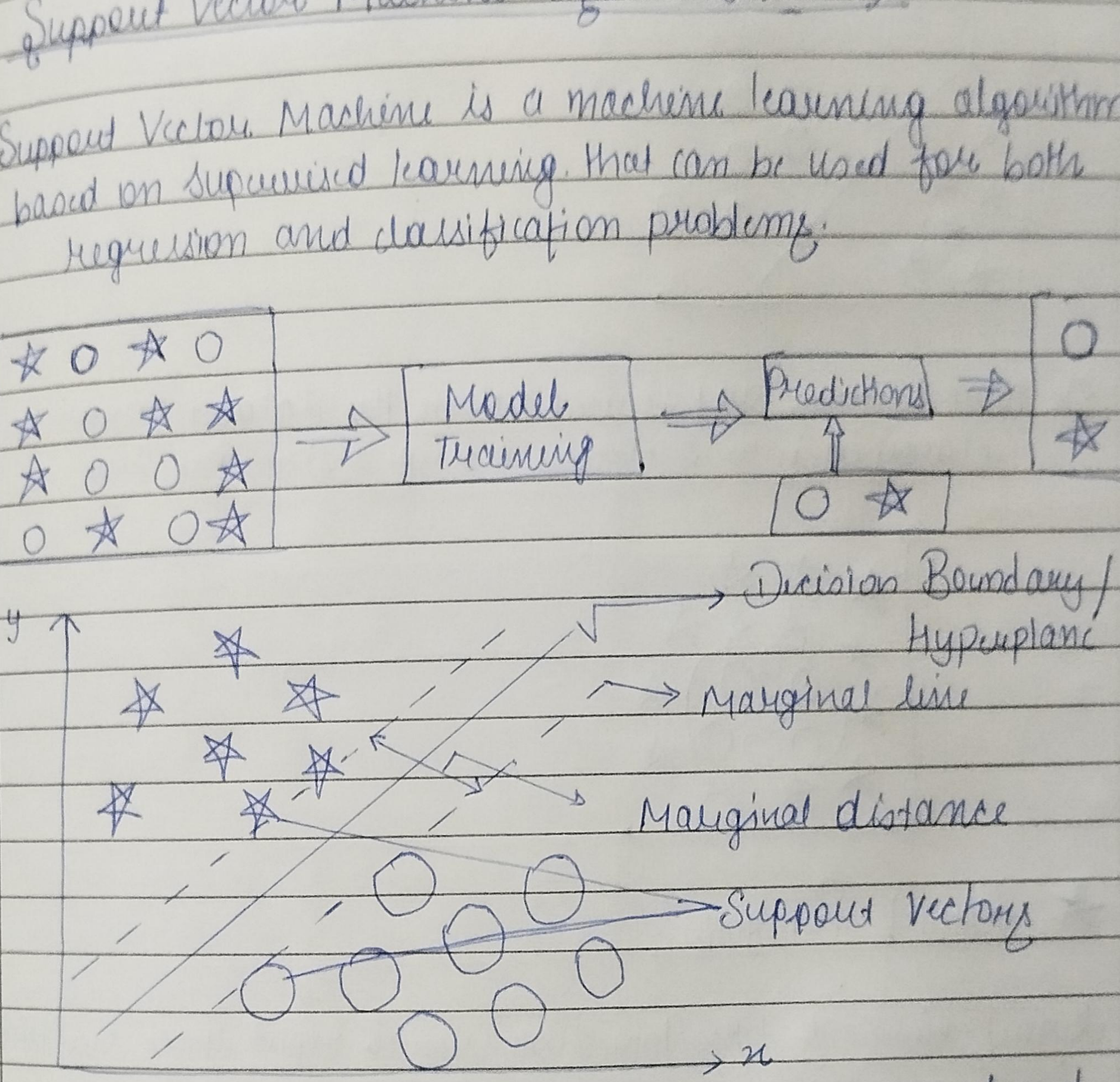
★ Support Vector Machine Algorithm (SVM)

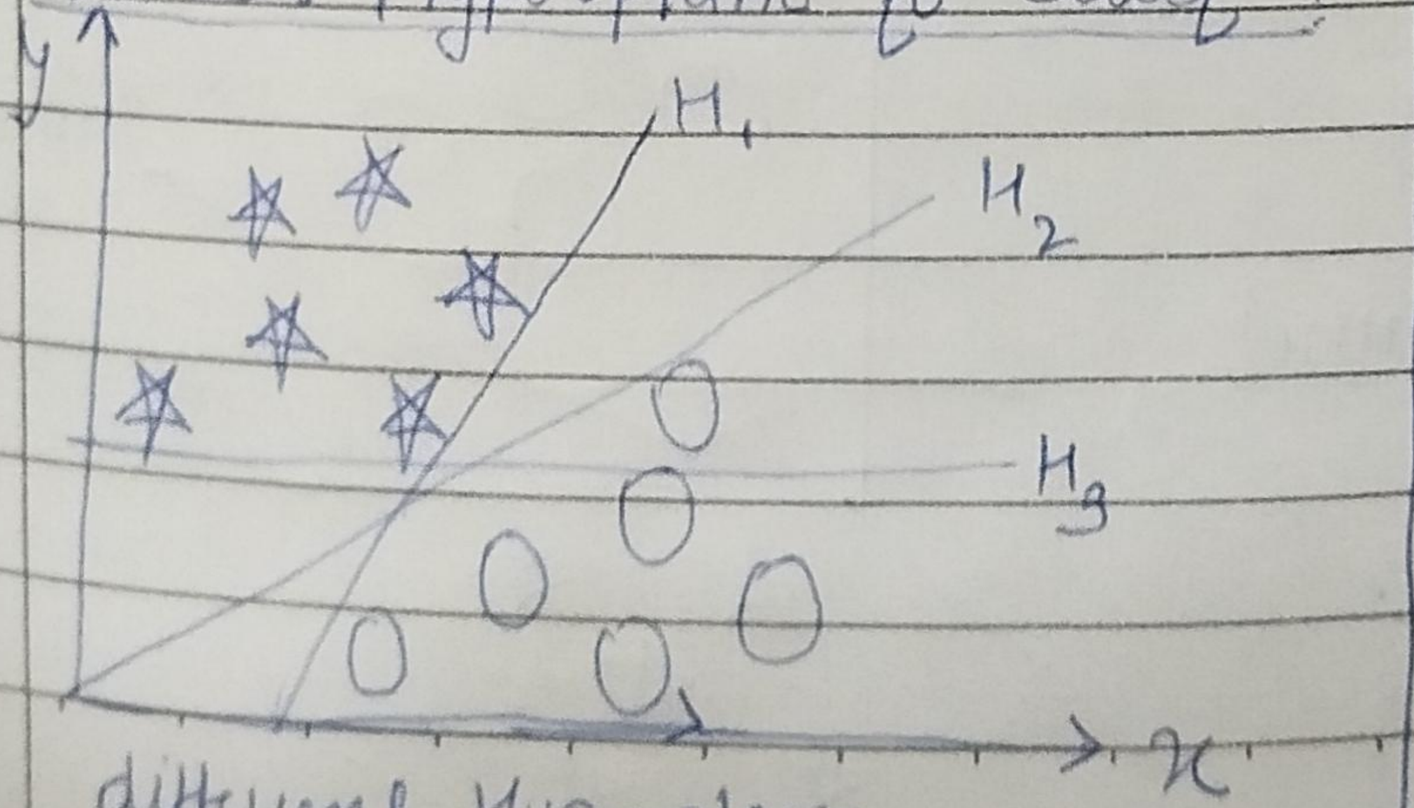
Support Vector Machine is a machine learning algorithm based on supervised learning that can be used for both regression and classification problems.





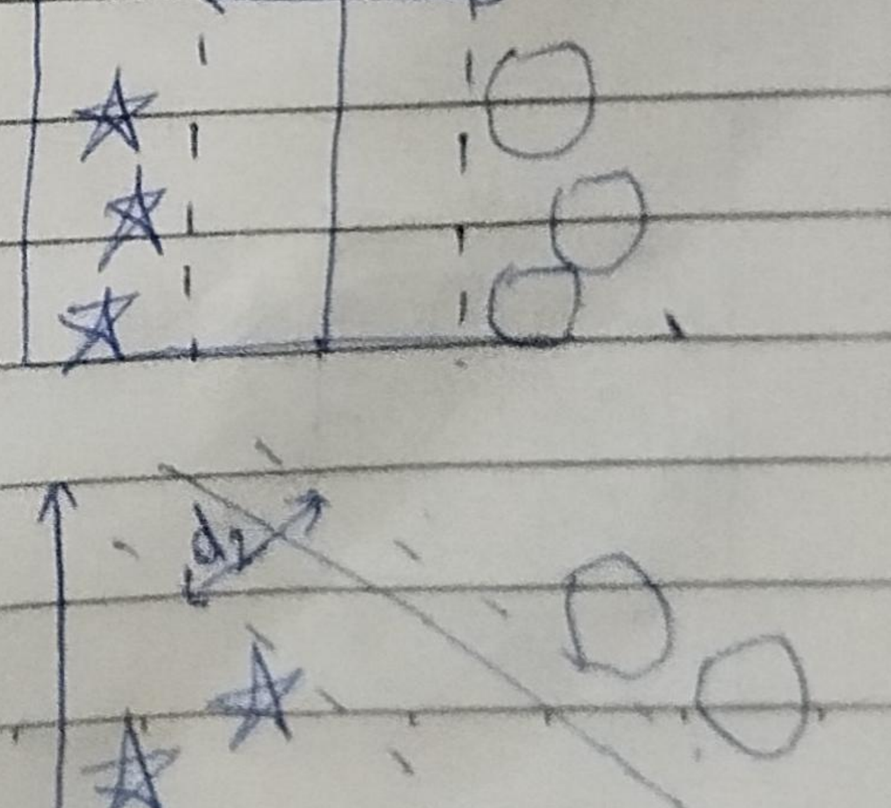
de > d1

★ Which Hyperplane to Select ?



different Hyperplane

H1, H2, H3

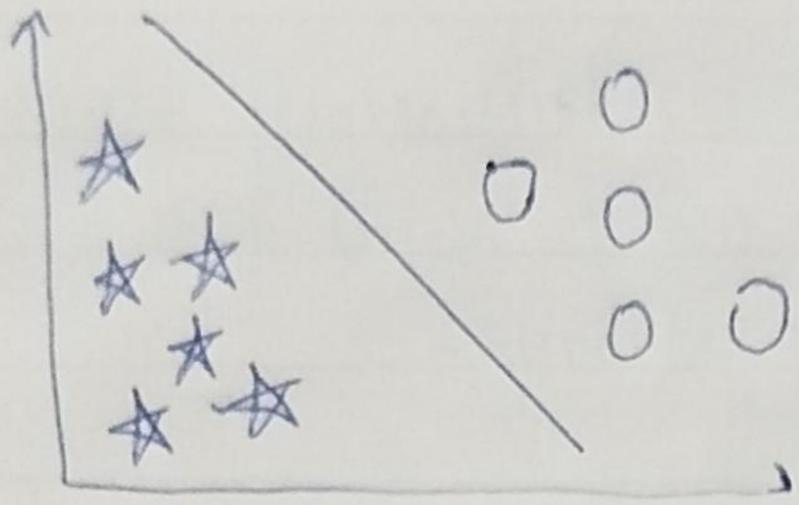


Maximum margin hyperpl

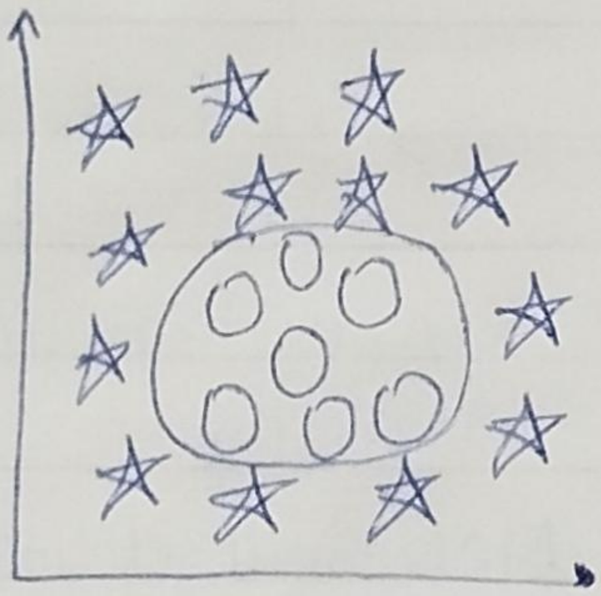
Teacher's Signature _____

★ Types of SVM

1) Linear SVM is used when dataset can be classified into 2 classes using a straight line.

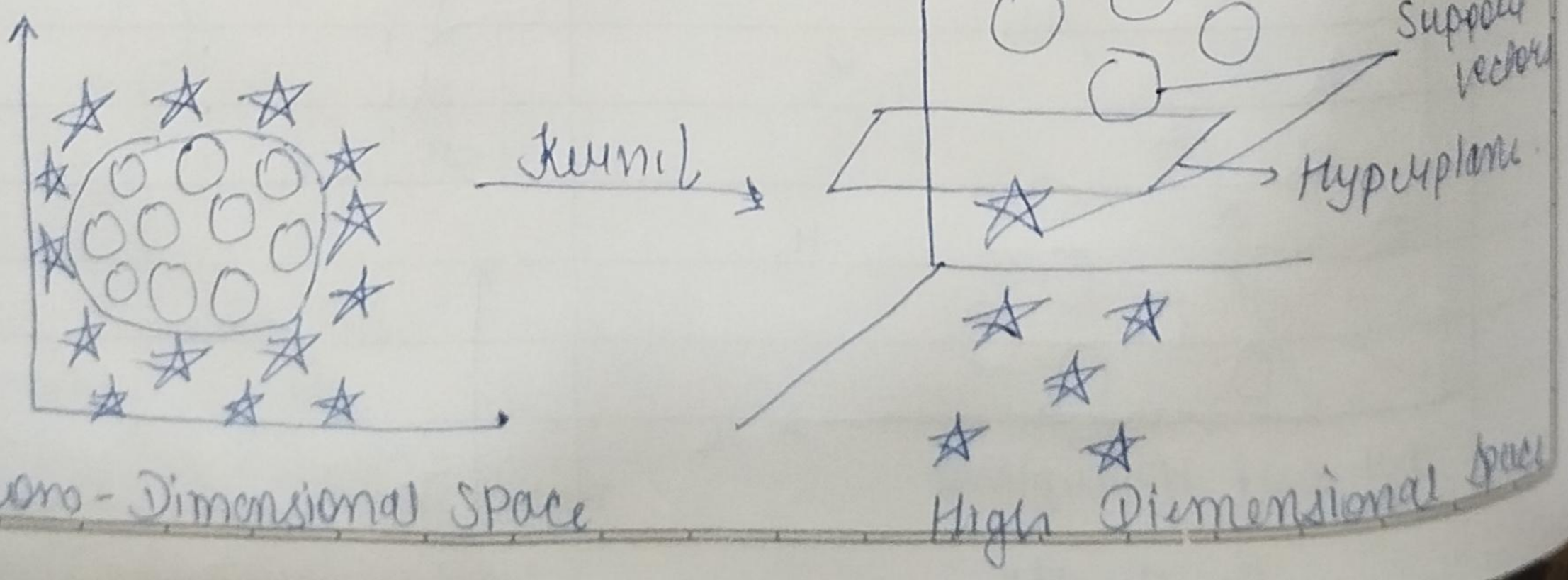


2) Non-linear SVM is used when the dataset cannot be classified into 2 classes using a straight line.



★ Kernel Function

Kernel Functions takes low dimensional input space and transforms it into a higher-dimensional space, i.e, it converts not separable problem to separable problem.



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Code

```
import pandas as pd
data = pd.read_csv(r"C:\Users\Singh\Downloads\iris-dataset.csv")
data
data
x=data.iloc[:,0:4]
x
y=data.iloc[:,5]
y

from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.2)
len(x_train)
len(x_test)

from sklearn.svm import SVC
model=SVC(kernel='linear')
model.fit(x_train,y_train)

model.predict(x_test)
x_test
model.score(x_test,y_test)
```

Teacher's Signature _____